LX200V50 WisLink 1000 Mbps PLC Module + EVB Datasheet

Overview

Description

LX200V50 is a Power Line Communication (PLC) module based on Qualcomm QCA7550 System-on-Chip (SoC). QCA7550 is a MAC/PHY transceiver module that meets the HPAV2 standard. It is designed to bridge multi-stream Ethernet content from a powerline network to an Ethernet 802.3 network. The PLC's specific MAC manages network admission and service flows to maximize the Quality of Service (QoS) over the powerline network.

Features

- HomePlug® AV2 compliant:
  - AV2 30 MHz and 67 MHz MIMO profiles
  - AV2 30 MHz and 67 MHz SISO profiles
  - Power Save Mode
- IEEE 1901, IEEE 1905.1, and HomePlug AV compliant
- Coexistence with HomePlug 1.0/Turbo nodes
- Integrated MAC/PHY, analog front end, and line driver
- Supports RGMII/RMII Ethernet with both MAC and PHY mode
- Internal DDR2 16 MB memory
- HomePlug AV2 compliant PHY:
  - Supports OFDM 4096/1024/256/64/16/8-QAM, QPSK, BPSK, ROBO, HS-ROBO, Mini-ROBO
  - Supports Line-Neutral/Line-Ground 2×2 MIMO with beam-forming
  - 128-bit AES Link Encryption with key management

- Windowed OFDM with noise mitigation based on patented line synchronization techniques improves data integrity in noisy conditions
- Dynamic channel adaptation and channel estimation
- Advanced Turbo Code Forward Error Correction (FEC)
- Turbo Decoder Rates: 1/2, 16/21, 8/9

- HomePlug AV2 Compliant MAC:
  - Priority-based CSMA/CA channel access schemes maximize efficiency and throughput
  - Integrated Quality of Service (QoS) Enhancements
  - Hardware Packet Classifiers for Type of Service (ToS), Class of Service (CoS), and IP Port Number
  - Supports IGMP managed multicast sessions

Specifications

Overview

Block Diagram

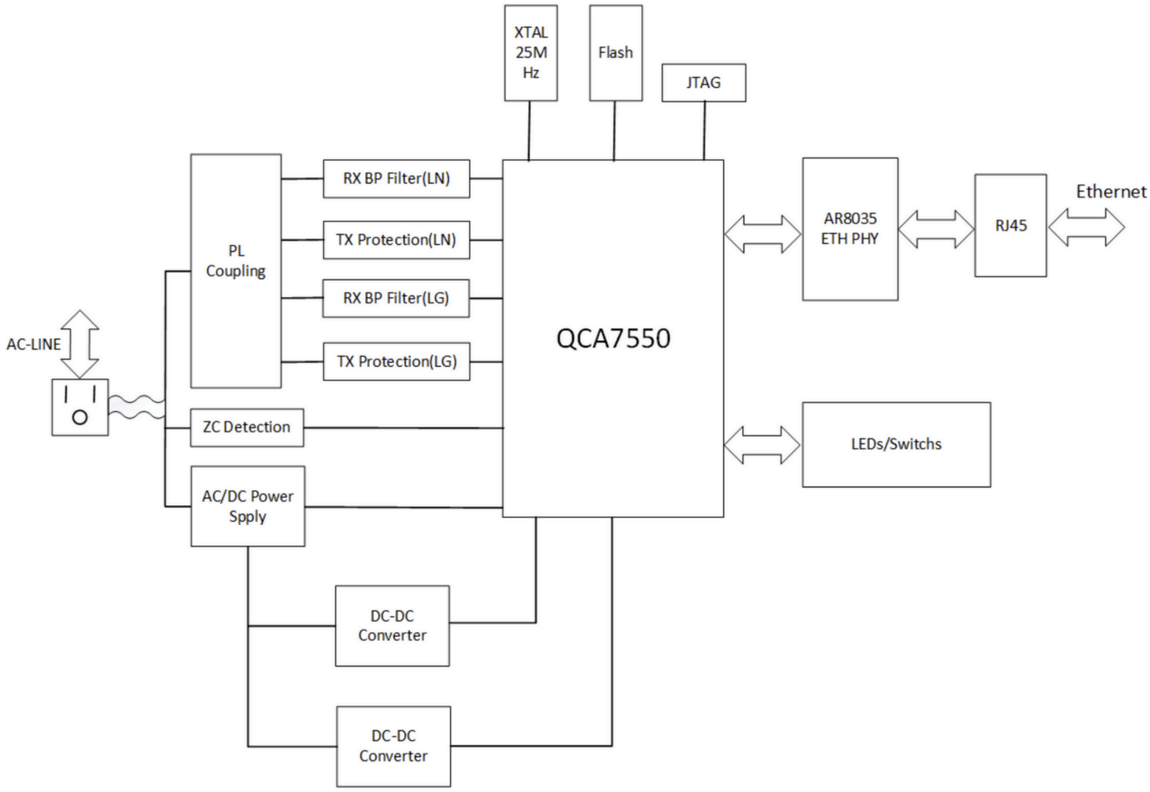

Figure 1: Block diagram

Parameters

Parameter	Value
Max. data rate	1000 Mbps
DC interface	4.75 V~36 V
V50 supported unit connection	6~8

Applications

- High Definition (HD) and Standard Definition (SD) video distribution
- Internet Protocol Television (IPTV) distribution
- Home networking backbone for Hy-Fi, Wi-Fi, and wireless USB
- High-speed broadband sharing

- Audio and video streaming and internet downloading
- Voice over Internet Protocol (VoIP)
- PC files and applications sharing
- Printer and peripheral sharing
- Network and interactive online gaming
- Security Hy-Fi cameras

Hardware

The hardware specification is categorized into five parts. It shows the interfacing and the pinouts of the board and their corresponding functions and diagrams. It also covers the electrical and mechanical parameters of the LX200V50 WisLink 1000 Mbps PLC Module + EVB.

Interfaces

Evaluation Board (EVB)

The EVB can be used for the customer's product development and testing, and can also be used as a part of the research and development of products. The baseboard is designed for all the external interface circuits required by the module, including coupling parts such as 3.3 V buck, Ethernet RJ45, PLC coupling transmission transformer, and other interfaces. It is mainly the interface. The structure of each product is different. The auxiliary interface evaluation board is designed to be convenient for testing.

Figure 2 shows the interfaces of the evaluation board.

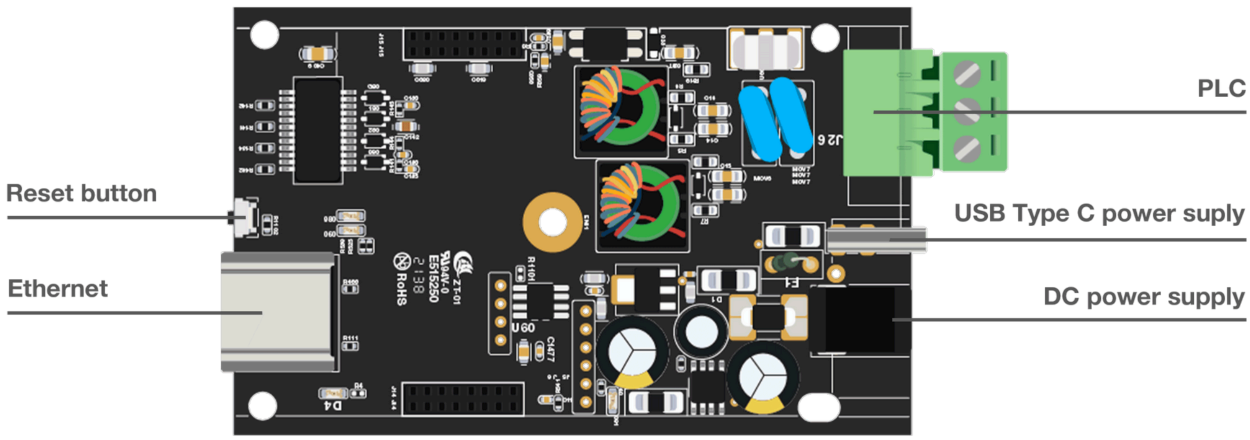

Figure 2: Baseplate interfaces and top view

Pin Definition

Pin	Description
NC	No connect
I	Input
I-	Differential input -
I+	Differential input +
I/O	Input/Output
O	Output
O-	Differential output -
O+	Differential output +
PI	Power input

J13 Pin Definition

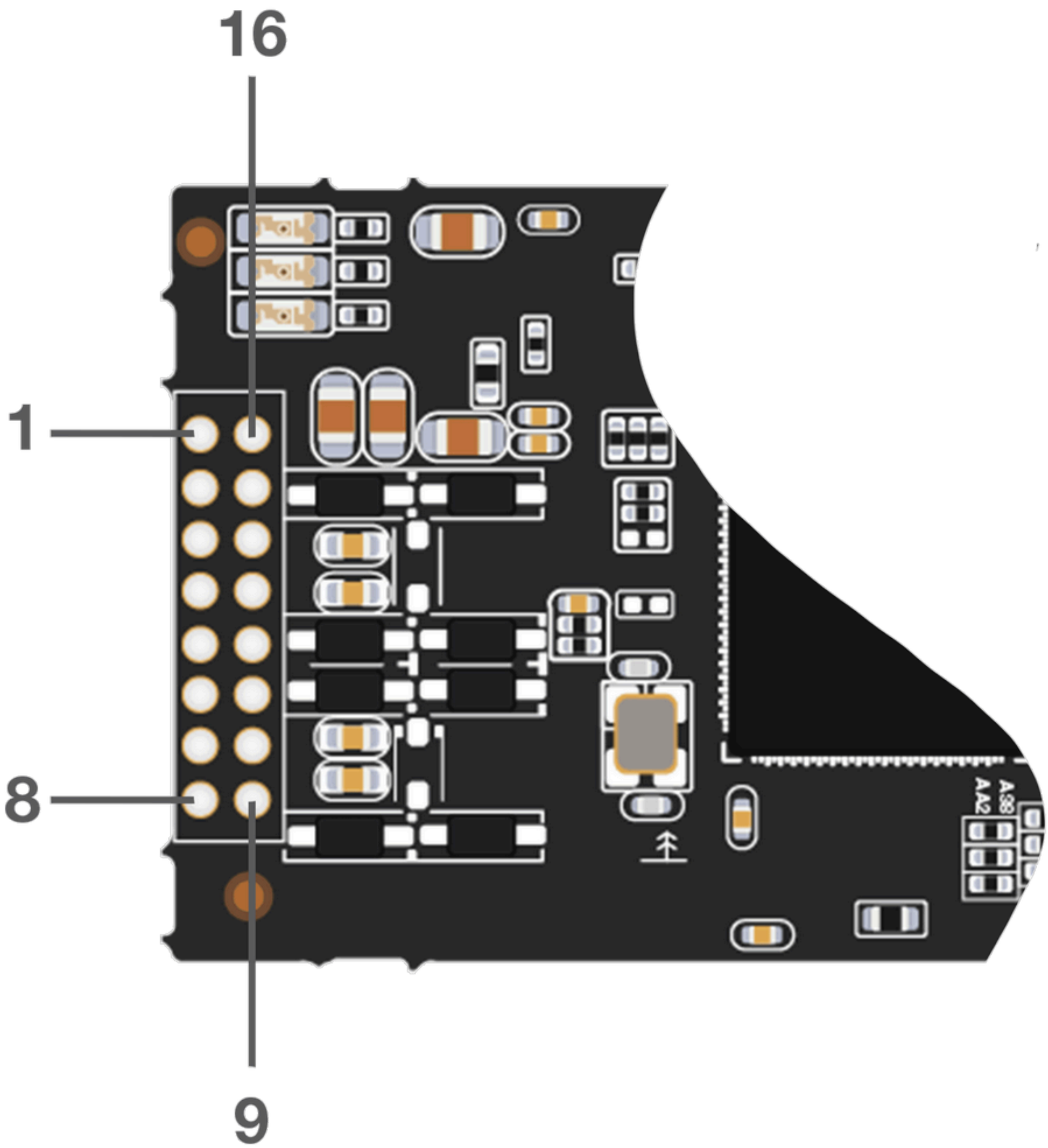

Figure 3: J13 pin order

Pin No.	Name	Direction	Description
1	ZC_INP	I	Zero cross detection

Pin No.	Name	Direction	Description
2	-	NC	Do not connect anything
3	-	NC	Do not connect anything
4	GND	-	Ground connection
5	GND	-	Ground connection
6	-	NC	Do not connect anything
7	-	NC	Do not connect anything
8	-	NC	Do not connect anything
9	TX1-	O-	MIMO Differential output for MIMO Channel #1
10	TX1+	O+	MIMO Differential output for MIMO Channel #1
11	RX1-	I-	MIMO Differential input for MIMO Channel #1
12	RX1+	I+	MIMO Differential input for MIMO Channel #1
13	TX0-	O	MIMO Differential output for MIMO Channel #0
14	TX0+	O	MIMO Differential output for MIMO Channel #0
15	RX0-	I	MIMO Differential input for MIMO Channel #0
16	RXO+	I	MIMO Differential input for MIMO Channel #0

J14 Pin Definition

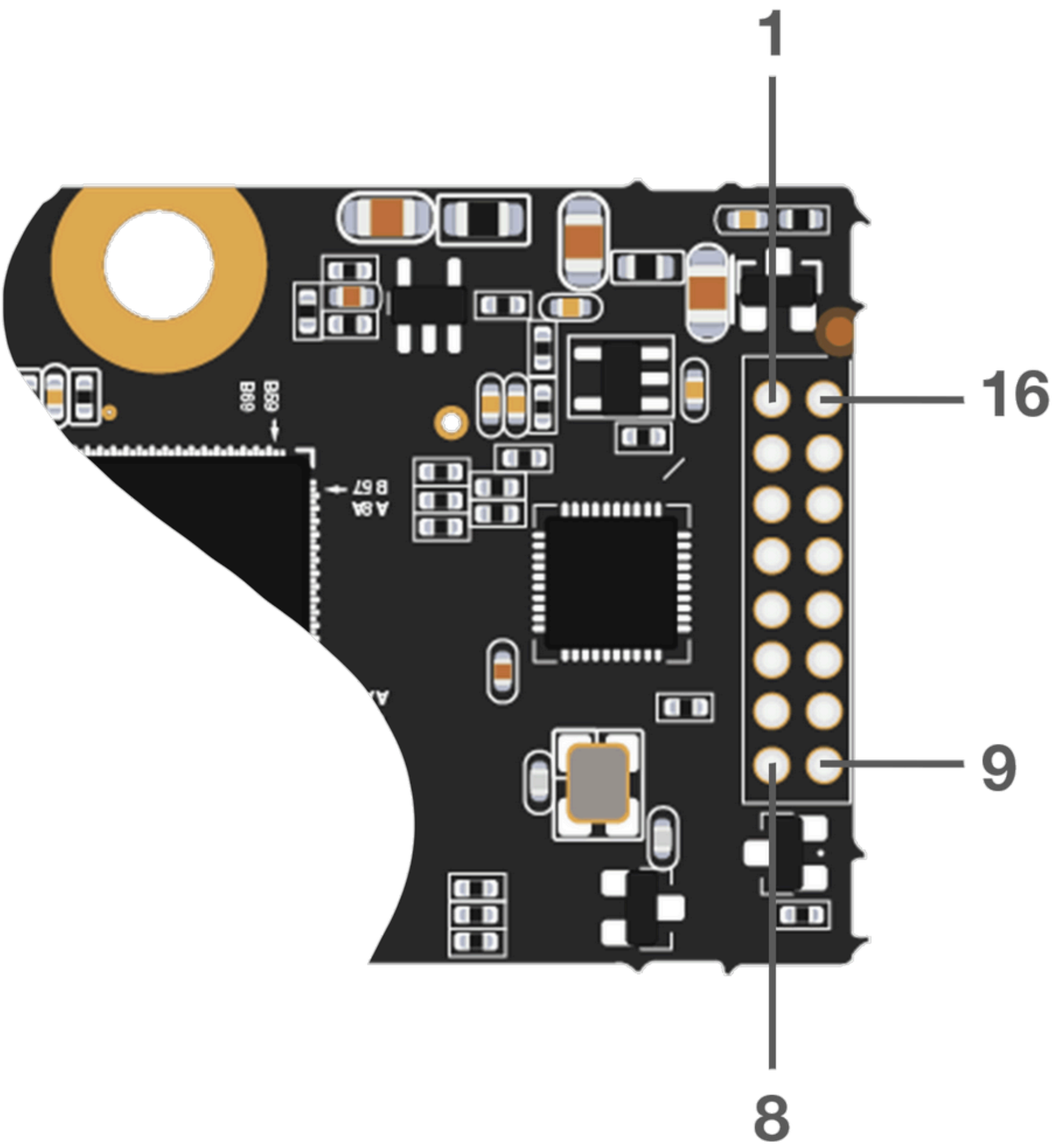

Figure 4: J14 pin order

Pin No.	Name	Direction	Description
1	TD3-	I/O-	Media-dependent interface 3, 100 Ω transmission line
2	TD3+	I/O+	Media-dependent interface 3, 100 Ω transmission line

Pin No.	Name	Direction	Description
3	TD2-	I/O-	Media-dependent interface 2, 100 Ω transmission line
4	TD2+	I/O+	Media-dependent interface 2, 100 Ω transmission line
5	TD1-	I/O-	Media-dependent interface 1, 100 Ω transmission line
6	TD1+	I/O+	Media-dependent interface 1, 100 Ω transmission line
7	TD0-	I/O-	Media-dependent interface 0, 100 Ω transmission line
8	TD0+	I/O+	Media-dependent interface 0, 100 Ω transmission line
9	3.3VPS	PI	Power supply
10	3.3VPS	PI	Power supply
11	GND	-	Ground connection
12	GND	-	Ground connection
13	GPIO[8]	IO	General purpose I/O, Power LED Blue
14	GPIO[6]	IO	General purpose I/O, LP LED Activity Green
15	GPIO[10]	IO	General Purpose I/O, Power LED Red
16	RESET_L	I	QCA7550 reset (low active)

Electrical Characteristics

Absolute Maximum Ratings

Parameter	Symbol	Min.	Typ.	Max.	Unit
Supply input voltage	3.3VPS	-2.7	3.3	5.5	V
Storage temperature	Tstore	-40	25	150	° C

Operation and Storage Temperatures

Parameter	Symbol	Min.	Typ.	Max.	Unit
Storage temperature	Tstore	-40	25	150	° C
Operating temperature	Ta	-40	25	80	° C

DC Characteristics

Parameter	Symbol	Min.	Max.	Unit
Low-level input voltage	VIL	-0.3	0.7	V
High-level input voltage	VIH	2.0	3.5	V
Low-level output voltage	VOL	-0.3	0.4	V
High-level output voltage	VOH	2.4	-	V
GPIO low-level current	IOGPIO	-12	12	mA

Mechanical Characteristics

All dimensions are measured in mm. The tolerance value is ±0.05 mm.

Module Dimensions

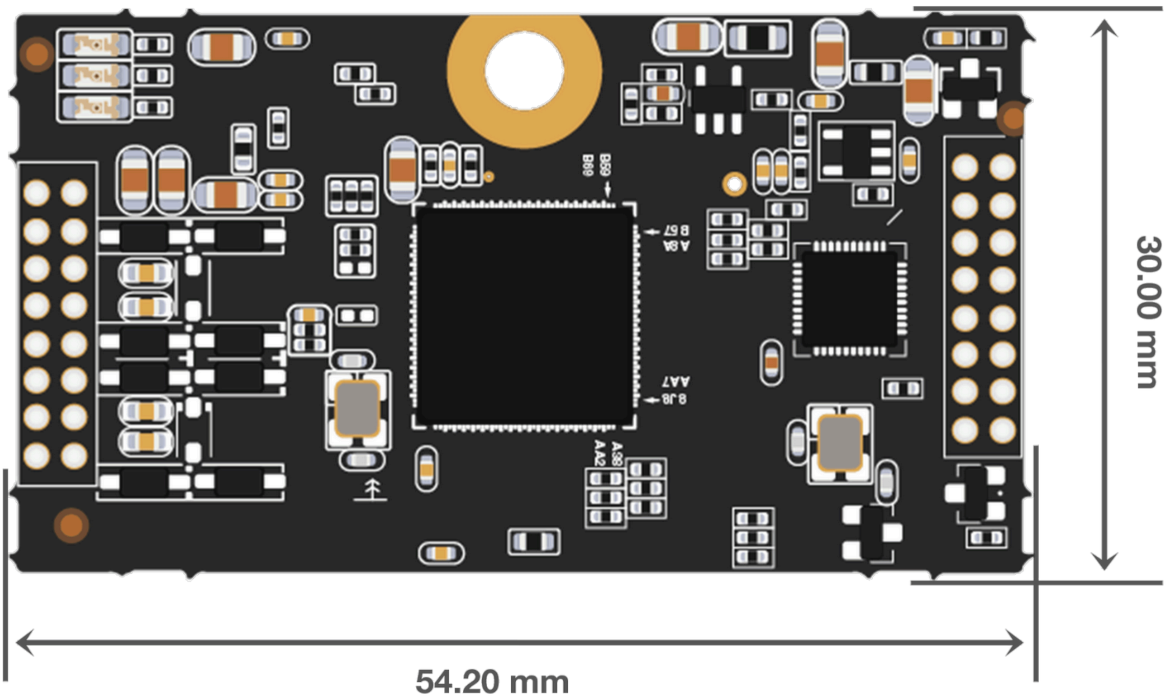

Figure 5: Module dimensions

Pin Dimensions

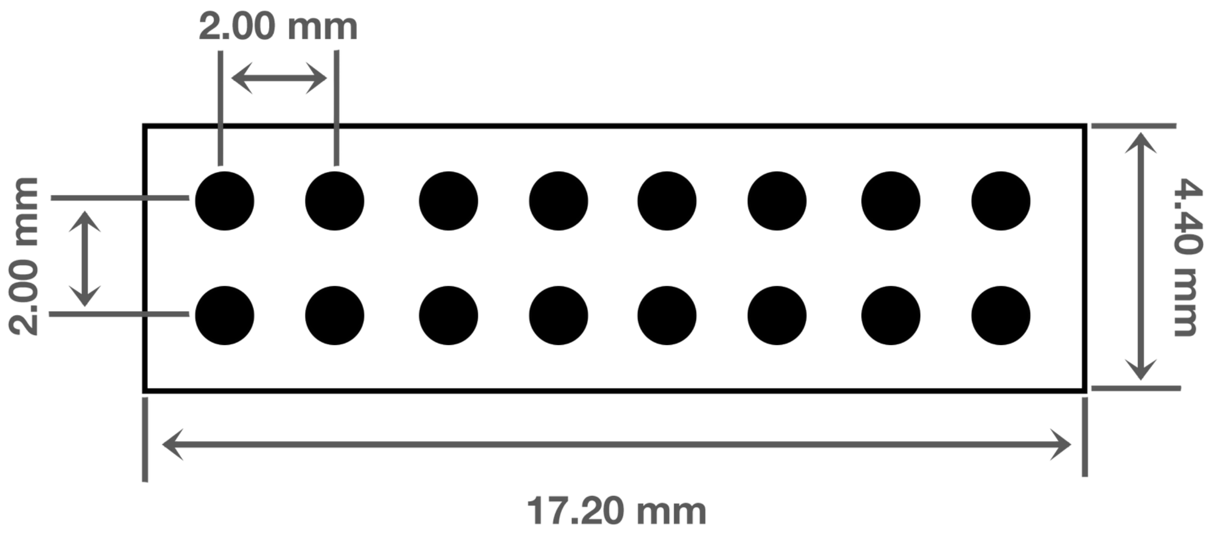

Figure 6: Pin dimensions

Baseplate Dimensions

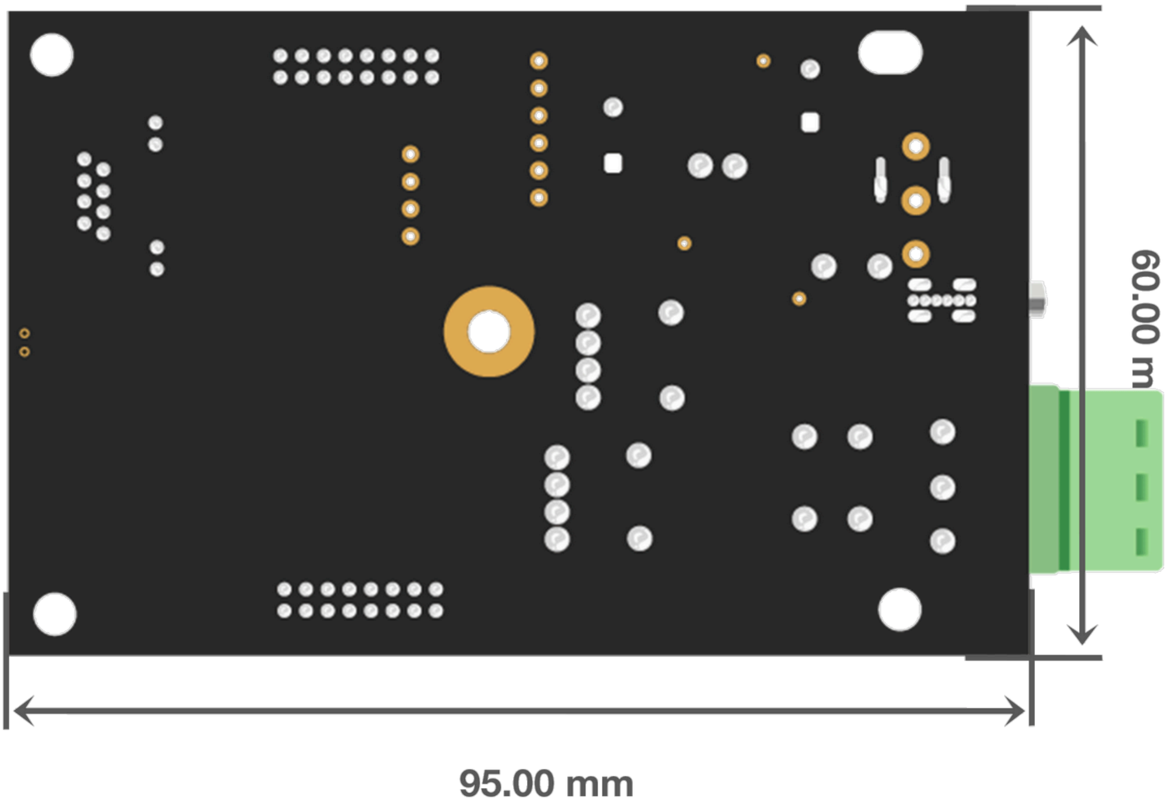

Figure 7: Baseplate dimensions and bottom view

Schematic Diagram

Power Supply

The board provides two power supply modes: DC power interface and Micro USB interface. The voltage input range of the DC interface is 4.75~36 V. The power supply is converted to the 3.3 V voltage required by the module through DC-DC. **Figure 8** shows the schematic diagram of the EVB's power supply.

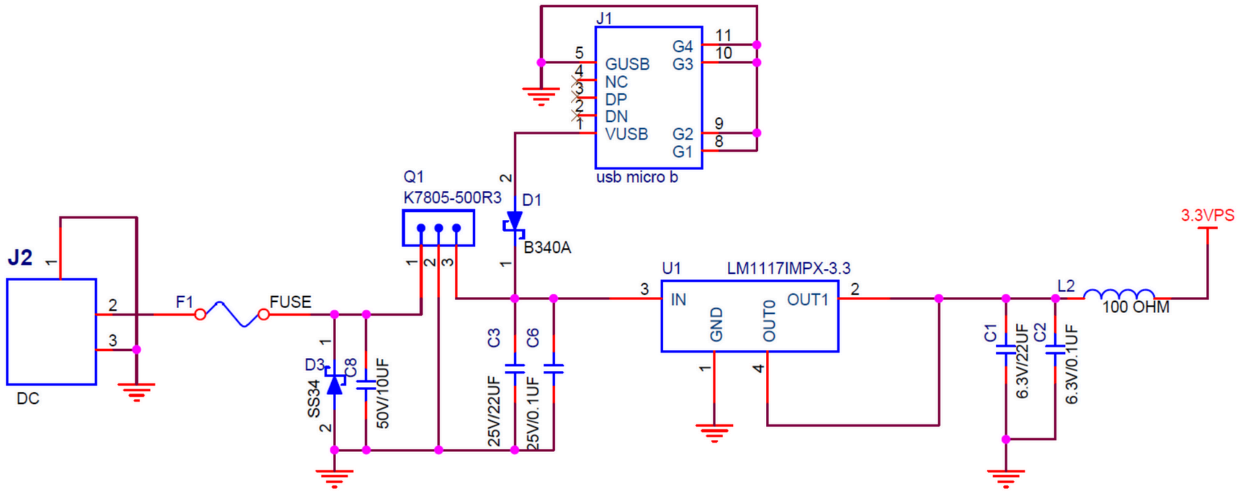

Figure 8: Power supply schematics

Ethernet and LEDs

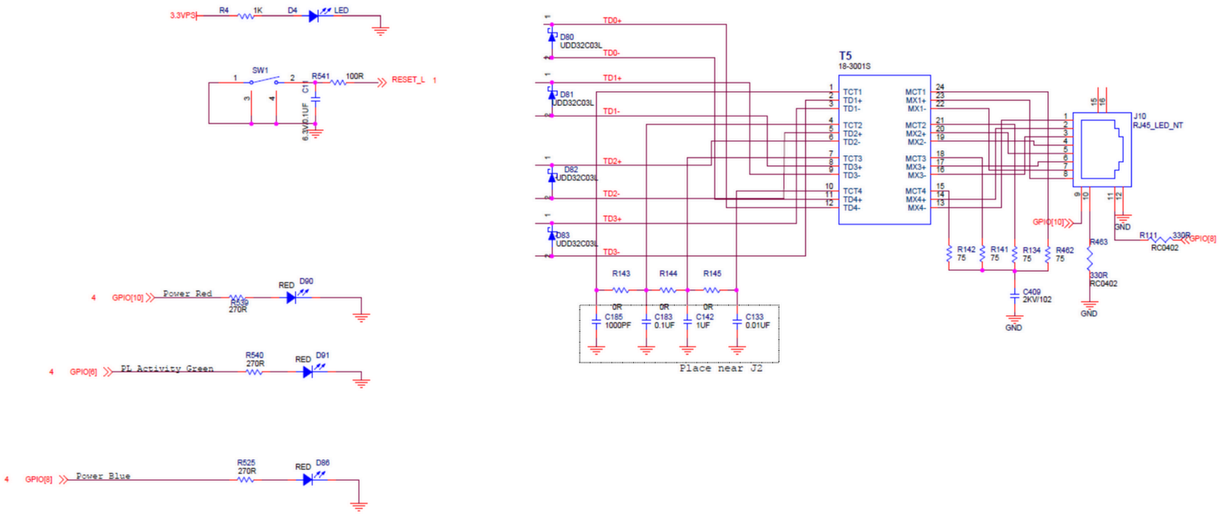

Figure 9: Ethernet and LEDs

Certification

CE FC

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